# Supplementary Information for: Hard thermodynamic constraints and the temperature extrapolation of learned water isotherms in metal-organic and covalent organic frameworks: a pre-registered test

Abdulhamid Batayhi

**Supplementary Table S1: Data**

| Material | Class | Set | Data | Source DOI | $T$ range (K) | $n_T$ | Points per $T$ | $p/p_{\mathrm{sat}}$ range | $q_{\mathrm{ref}}$ (mol kg$^{-1}$) |
|---|---|---|---|---|---|---|---|---|---|
| COF-432 | COF | conf. | vector | 10.1021/jacs.9b13094 | 283.15–313.15 | 3 | 28–29 | 0.063–0.870 | 16.04 |
| COF-482-hydrazide | COF | conf. | raster | 10.1021/acscentsci.2c00398 | 288.15–308.15 | 3 | 25–51 | 0.08–0.898 | 34.70 |
| COF-SO3H | COF | conf. | vector | 10.1002/cssc.202301906 | 298.15–318.15 | 3 | 18–18 | 0.0003–0.900 | 17.04 |
| HFPTP-DMePDA-COF | COF | conf. | raster | 10.1038/s41467-021-27128-4 | 283.15–298.15 | 3 | 9–18 | 0.39–0.897 | 7.91 |
| HFPTP-PDA-COF | COF | conf. | raster | 10.1038/s41467-021-27128-4 | 283.15–298.15 | 3 | 24–32 | 0.0041–0.898 | 16.78 |
| NO-PI-3-COF | COF | conf. | vector | 10.1021/jacs.3c02572 | 288.15–308.15 | 3 | 30–37 | 0.0091–0.882 | 12.74 |
| PI-3-COF | COF | conf. | vector | 10.1021/jacs.3c02572 | 288.15–308.15 | 3 | 28–37 | 0.01–0.887 | 24.80 |
| TFBCz-PDA-COF | COF | conf. | raster | 10.1038/s41467-021-27128-4 | 283.15–298.15 | 3 | 29–30 | 0.0023–0.899 | 27.26 |
| Al-fumarate | MOF | conf. | raster | 10.1021/acscentsci.9b00745 | 283.15–323.15 | 3 | 56–59 | 0.049–0.892 | 23.86 |
| CALF-25 | MOF | conf. | ISODB | 10.1021/ja306812r | 298.00–313.00 | 3 | 17–20 | 0.018–0.894 | 3.75 |
| Co2Cl2(BTDD) | MOF | conf. | vector | 10.1021/acscentsci.7b00186 | 283.00–298.00 | 3 | 31–34 | 0.0072–0.862 | 53.97 |
| Mg-SU-102 | MOF | conf. | numeric | 10.1038/s41467-025-59551-2 | 288.15–328.15 | 9 | 21–21 | 0.0013–0.819 | 21.11 |
| Zn2(bptc) | MOF | conf. | ISODB | 10.1021/ja060946u | 288.00–318.00 | 4 | 25–33 | 0.023–0.885 | 8.38 |
| TFPPy-PDA-COF | COF | dev. | raster | 10.1038/s41467-021-27128-4 | 283.15–298.15 | 3 | 7–15 | 0.46–0.900 | 35.41 |
| CuBTC | MOF | dev. | ISODB | 10.1016/j.ijheatmasstransfer.2012.07.068 | 293.00–358.00 | 9 | 23–41 | 0.0029–0.508 | 25.29 |
| MIL-100 | MOF | dev. | ISODB | 10.1016/j.ijheatmasstransfer.2012.07.068 | 293.00–358.00 | 9 | 25–35 | 0.011–0.603 | 18.11 |
| MIL-160 | MOF | dev. | ISODB | 10.1002/adma.201502418 | 293.00–313.00 | 3 | 32–54 | 0.0052–0.884 | 21.79 |

Supplementary Table S1. The 13 confirmatory materials (conf.) and the four development materials (dev.) after the inclusion rules and the preprocessing. Data: how the points were obtained (vector or raster digitisation of the published figure, numeric source data, or the NIST adsorption database, ISODB). $n_T$: number of temperatures; points per temperature with $p/p_{\mathrm{sat}} \leq 0.90$; $q_{\mathrm{ref}}$: the largest uptake over all temperatures in this range, used only for scoring. Every decision and its reason is in data/inclusion_log.csv.

## Supplementary Note 1: Can each verdict fail? Synthetic worlds and power

Before the freeze, the whole pipeline (the tuned configurations, fitting from 5 seeds, the violation counts and the analysis code) was run in synthetic worlds that share no uptake data with the confirmatory set. Each world has 13 materials; material $i$ uses the temperatures of confirmatory material $i$. An expected outcome was written for each world before it ran. In KS0 the truth is a two-site Sips isotherm with van't Hoff affinities, inside the family of P; in DA it is a Dubinin-Astakhov curve, inside the family of the hard characteristic curve (A-hard); in FW it is a Kirchhoff-Sips isotherm with a large heat capacity, so that van't Hoff is wrong; FLAT has one weakly curved Langmuir site; SUB has a sub-linear Sips isotherm outside the family of P and inside that of the penalty model V-pen.

The comparison of P with A-hard discriminated in both directions: P extrapolated better in KS0 ($R = 0.52$, exact 95 % interval 0.35 to 0.70) and A-hard better in DA ($R = 1.26$, 1.07 to 1.49). Three expectations were missed. In FW the comparison of P with the free temperature network F-hard still read "P better" ($R = 0.33$): with two training temperatures the free network cannot extrapolate, so this comparison cannot fail and is reported descriptively. In FLAT the unconstrained network F-none violated physics "often". In SUB, built for V-pen to win, P was slightly better ($R = 0.96$, 0.94 to 0.99); the mixture of Langmuir terms in P approximates a sub-linear isotherm over the measured range. The accuracy comparison of hard constraints with penalties therefore could not be made to fail synthetically, and it is tested as non-inferiority.

The power of each verdict was simulated with the frozen analysis code at $n = 13$ (2000 simulations per setting). At the spread expected for the comparison with V-pen (0.158 decades), the non-inferiority criterion is met with probability 0.0225 when the true ratio is at the margin, $R = 1.25$, and 0.51 (normal) or 0.74 (Student $t$, 3 degrees of freedom) when the two models are equally accurate. A "not met" outcome is therefore likely even under equal accuracy and is never read as evidence that the constraints cost accuracy.

## Supplementary Note 2: Synthetic truths for the isosteric heat

The heat test uses five truth families with exact heats. Their shape parameters are fitted by bounded least squares to each confirmatory material's middle-temperature isotherm; their thermodynamic parameters are fixed in advance: KS, a Langmuir site with $Q_0 = 55$ kJ mol$^{-1}$ plus a cooperative Sips site with $Q_0 = 48$ kJ mol$^{-1}$, a heat capacity of 40 J mol$^{-1}$ K$^{-1}$ and an exponent of at least 1.5, inside the family of the Kirchhoff variant of P and near that of P; DA, a Dubinin-Astakhov curve $q = W_0 \exp[-(A/E)^n]$ with $1 \le n \le 6$, inside the family of the characteristic curve; FG, a Fowler-Guggenheim isotherm with $Q_0 = 45$ kJ mol$^{-1}$ and a lateral attraction between 0 and $3.9\,RT_{\min}$, inside neither; HS, strong hydrophilic sites, a Langmuir site with $Q = 80$ kJ mol$^{-1}$ holding at most 0.3 $q_{\mathrm{ref}}$ plus a Sips site with $Q = 46$ kJ mol$^{-1}$, inside the family of P with a heat that falls steeply with loading; TC, a Sips isotherm with $Q = 50$ kJ mol$^{-1}$ whose capacity falls by 0.2 % per kelvin, inside the family of P and outside that of the characteristic curve. Gaussian noise of 0.4 % of $q_{\mathrm{ref}}$ is added with a fixed seed per material and family.